

Customer Shopping Behavior Analysis

A data analysis project exploring transactional patterns across **3,900 purchases** — using Python, SQL, and Power BI to surface actionable business insights.

Python

Data cleaning, feature engineering & EDA with pandas

PostgreSQL

Advanced SQL queries for revenue, segmentation & behavior

Power BI

Executive-ready dashboard with dynamic filtering

Project Overview & Dataset

Objective

Uncover spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions across marketing, pricing, and retention.

Tech Stack

Python (pandas) → PostgreSQL → Power BI

Dataset at a Glance

Attribute	Detail
Dataset size	3,900 rows × 18 columns
Demographics	Age, Gender, Location
Purchase data	Item, Category, Amount, Season
Behavior signals	Subscription Status, Discount Applied, Shipping Type
Satisfaction	Review Rating
Data quality issue	37 missing values in Review Rating — imputed via median per category

Data Preparation in Python

The raw dataset was ingested, cleaned, and engineered using **pandas** before loading into PostgreSQL for SQL-layer analysis.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

Exploration

Imported dataset with pandas; used `df.info()` and `.describe()` to assess structure and summary statistics.

Missing Data

Imputed 37 null `review_rating` values using median rating per product category — preserving distributional integrity.

Standardization

Renamed all columns to `snake_case` for consistency across Python, SQL, and documentation.

Feature Engineering

Created `age_group` (binned ages) and `purchase_frequency_days` columns to enable richer segmentation queries.

Redundancy Check

Verified `discount_applied` and `promo_code_used` were redundant — dropped `promo_code_used` to reduce noise.

DB Integration

Connected Python script to PostgreSQL and loaded the cleaned DataFrame directly into the database for SQL analysis.

Revenue & Spending Analysis (SQL)

Key Findings

Gender Revenue Gap

Male customers generated **\$157,890** vs. Female customers at **\$75,191** — a **2.1× gap** with significant implications for targeting strategy.

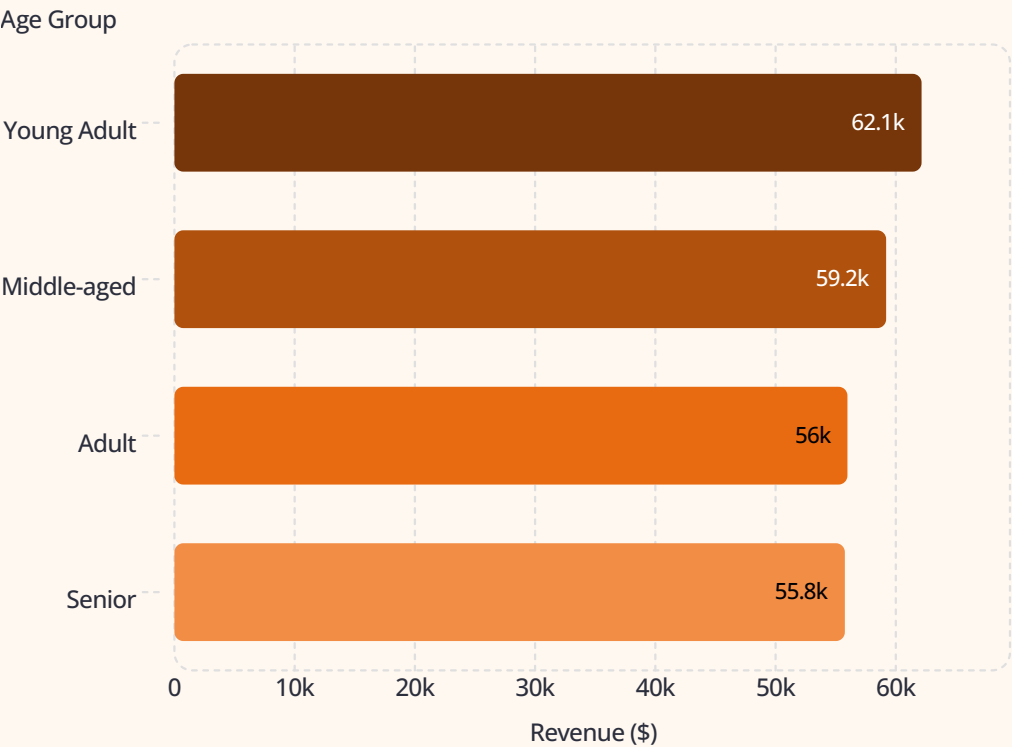
Shipping Type Signal

Express shipping avg. purchase (**\$60.48**) outpaces Standard (**\$58.46**) — higher-intent buyers self-select into faster delivery options.

High-Spend Discount Users

839 customers used discounts yet still spent above average — a high-value segment for precision-targeted promotions.

Revenue by Age Group



Young Adults lead revenue at **\$62,143**, with a steady decline across older cohorts — a clear signal for age-targeted spend allocation.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

Subscription & Loyalty Behavior

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

27%

Subscriber Share

Only 1,053 of 3,900 customers hold an active subscription

\$59.49

Subscriber Avg. Spend

vs. \$59.87 for non-subscribers — near-identical per-transaction value

\$170K

Non-Sub Revenue

Non-subscribers drive \$170,436 in total revenue vs. \$62,645 for subscribers

2,518

Repeat Non-Subscribers

Customers with 5+ purchases who have not converted to a subscription

⚠ Loyalty programs are not converting repeat buyers into subscribers. With 2,518 repeat non-subscribers, there is a significant untapped conversion opportunity.

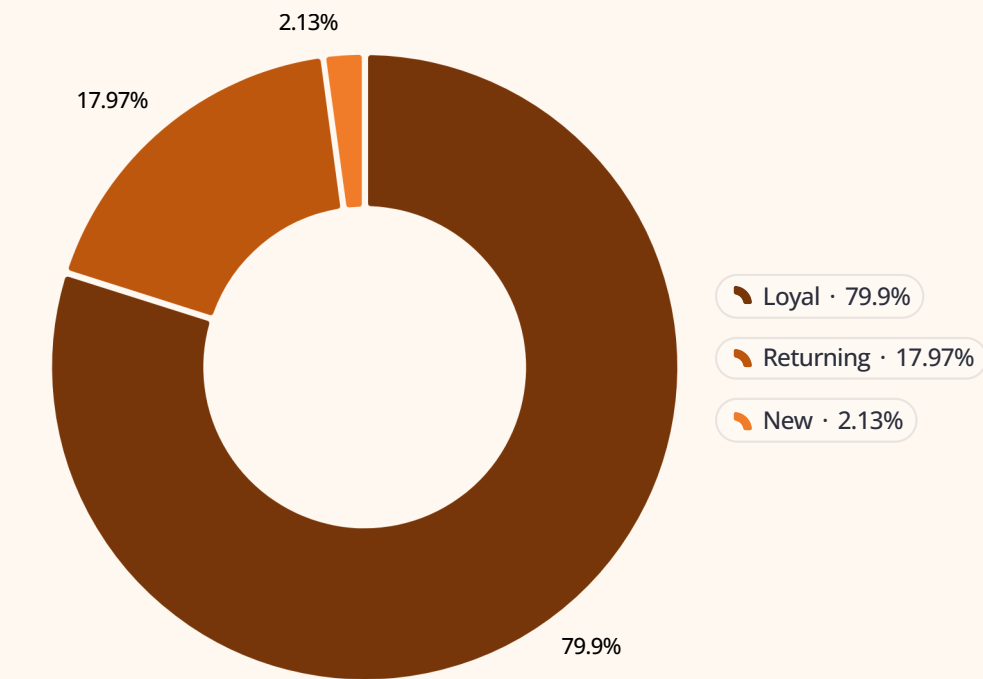
Customer Segmentation

Segmentation Logic

Customers were classified into three tiers based on purchase history count, defined entirely in SQL — no external ML tools required.

1	New First-time or very low purchase count — 83 customers. Critically low acquisition rate signals an urgent need for top-of-funnel investment.
2	Returning Mid-tier with repeat intent — 701 customers. Prime candidates for loyalty incentives to accelerate conversion into the Loyal tier.
3	Loyal Dominant segment — 3,116 customers (80%). Strong retention foundation, but concentration risk if acquisition remains stagnant.

Segment Distribution



	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

Product Performance & Discount Dependency

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

Top-Rated Products

Customer ratings are clustered in a narrow band — indicating broadly consistent satisfaction across the catalog.

Product	Avg. Rating
Gloves	3.86
Blouse	3.85
Jewelry	3.84
Pants	3.82
Skirt	3.78

Discount Dependency Risk

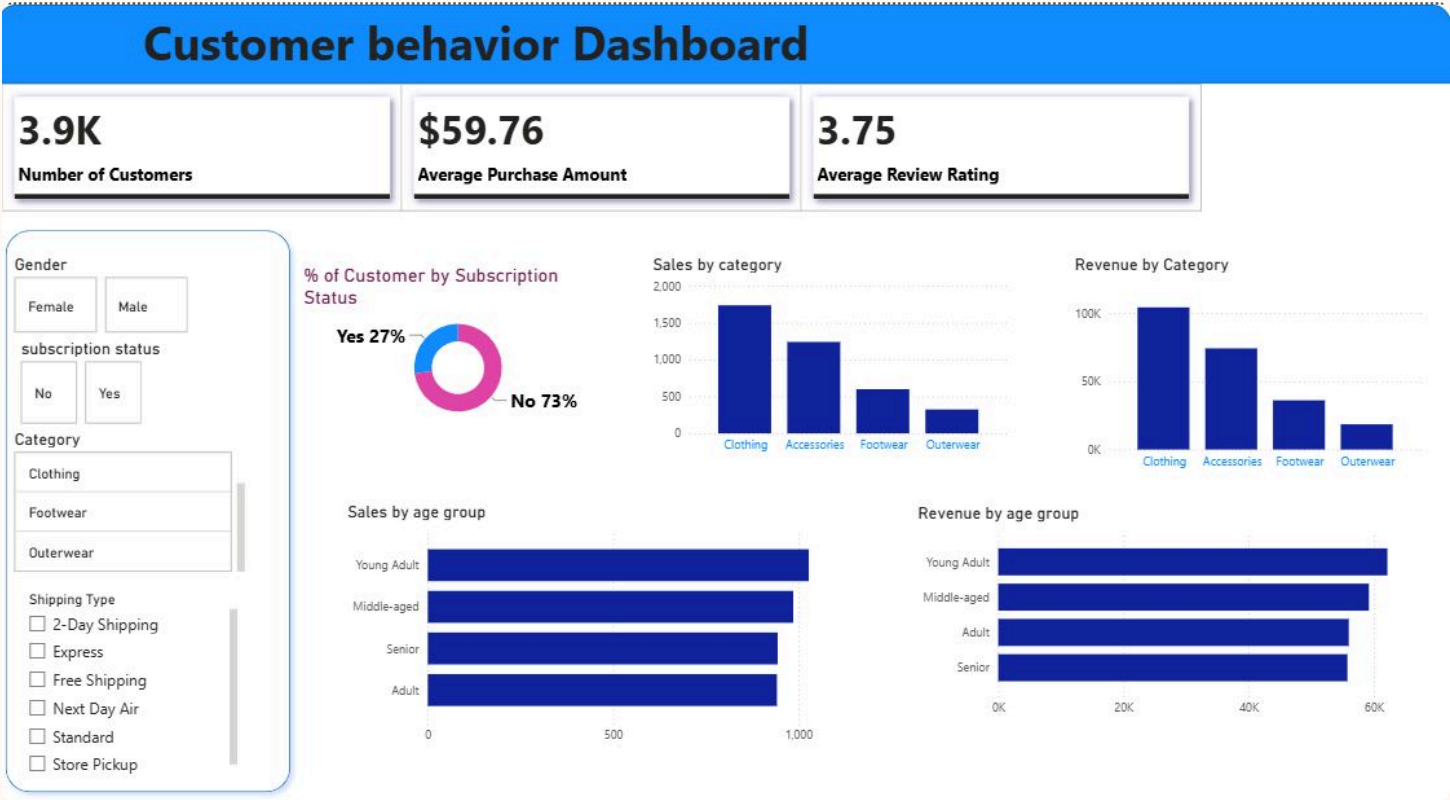
Several high-volume items are discounted nearly **50% of the time** — a margin erosion risk requiring a pricing strategy review.

Product	Discount Rate
Hat	50.0%
Sneakers	49.5%
Coat	49.0%
Sweater	48.0%
Pants	47.4%

⊗ High discount dependency on popular items risks significant margin erosion. A tiered or time-limited discount model should be evaluated immediately.

Power BI Dashboard

SQL query results were translated into an **executive-ready visual layer** — enabling non-technical stakeholders to interact with insights dynamically.



KPIs

3.9K customers · **\$59.76** avg. purchase · **3.75** avg. review rating

Visualizations

Pie charts for subscription status & category sales; bar charts for revenue by age group & product category

Interactivity

Filter panel for Gender, Subscription Status, and other attributes — enabling dynamic slicing of all insights

Business Recommendations

Boost Subscription Conversion

Only **27%** of customers subscribe. Promote exclusive benefits — early access, free shipping — to convert the **2,518 repeat non-subscribers** who already demonstrate loyalty behaviors.

Accelerate Returning → Loyal Migration

Reward the **701 Returning customers** with targeted loyalty incentives to accelerate their progression into the high-value Loyal segment.

Reform Discount Policy

Products like Hat and Sneakers are discounted **~50%** of the time. Introduce tiered or time-limited discounts to protect margins while preserving demand signals.

Elevate Top-Rated Products

Highlight top-rated items — **Gloves, Blouse, Jewelry** — and best-sellers by category in marketing campaigns to leverage existing customer satisfaction signals.

Target High-Value Segments

Concentrate acquisition spend on **Young Adults** (highest revenue: \$62,143) and **Express shipping users** (higher avg. spend: \$60.48) for maximum ROI.

Invest in New Customer Acquisition

With only **83 new customers** in the dataset, top-of-funnel campaigns are critical to diversify the customer base and reduce concentration risk in the Loyal tier.

Sources

| [Customer Shopping Behavior Analysis.docx](#)